S475/1
SUBSIDIARY
MATHEMATICS
PAPER 1
July/August 2026
$2\frac{1}{2}$ hours

WAKISSHA JOINT MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

SUBSIDIARY MATHEMATICS

PAPER 1

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

- *This examination paper consists of two sections* **A** *and* **B.** *It has* **six** *items.*
- *Section* **A** *has* **two** *compulsory items.*
- *Section* **B** *has* **two** *parts; part* **I** *and part* **II,** *each with two items. Respond to* **only two** *items from section B choosing* **one** *from each part.*
- *Respond to* **four** *items in all.*
- *Any additional items responded to will* **not** *be scored.*
- *All responses must be written in the answer booklets/sheets provided.*
- *Graph paper is provided.*
- *Silent non-programmable scientific calculators and mathematical tables with a list of formulae may be used.*

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SECTION A

All items in this section are **compulsory.**

Item I

Mr. Senkeezi is one of your neighbours. He would like to fence off his small piece of land on the side where members of the community trespass. One pole is at a point A with position vector $\overrightarrow{OA} = 6\underset{\sim}{i} + 2\underset{\sim}{j}$. While the second pole is at point B with position vector $\overrightarrow{OB} = 2\underset{\sim}{i} + 8\underset{\sim}{j}$. He has decided to add another pole at point C with position vector $\overrightarrow{OC} = 4\underset{\sim}{i} + 5\underset{\sim}{j}$ in order to reinforce the wire mesh. Unfortunately his neighbour claims that the third pole is not in line with A and B. He needs evidence to justify his action, otherwise he would have to relocate pole C.

Additionally, he plans to start a garage in his plot of land and one of the requirements is to construct an inclined plane for vehicle inspection.

The local builder he contacted has advised that the angle of incline should satisfy the equation $3\text{Cos}2\theta - \sin\theta + 2 = 0$. Mr. Senkeezi is aware that the angle of incline should be 30^0. He is not sure whether to go ahead and use the local builder, or find another alternative.

Task

Use numerical evidence to help Mr. Senkeezi make informed decisions on whether to;

(a) relocate pole C.

(b) use the local builder.

Item 2

A certain private school decided to buy a maize mill in order to cut food costs by milling their own grown maize. The output M (in tonnes) is modeled by;

$M = n^3 - 6n^2 + 9n + 2$, where n is the number of days the machine is operated per week. If the maximum output per week exceeds 5 tonnes, the school may consider selling part of the produce to the nearby community as another source of income.

The school started making monthly sales. It was noted that the sales(s) were directly proportional to twice the time(t) of production (in months). At the start of the business, one tonne of maize flour was sold. After 3 months, 5 tonnes were sold. If by the end of eight months less than 150 tonnes are sold, the school may consider widening the customer base beyond its immediate community.

Task

Basing of mathematical evidence help the school to make informed decisions.

SECTION B

PART I

Respond to **one** *item from this part.*

Item 3

Mr. Kaluma has to repay a loan of 3 million shillings. He has been given two repayment options:

Option 1:

He makes an initial deposit and keeps increasing the monthly installments by a constant amount such that;

In the 5th month he would pay Ugx. 90,000

In the 12th month he would pay Ugx. 160,000

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Option 2:

The repayment plan is modeled by $7^x = 12$, where x is the period of payment. His target is to repay the loan in the shortest time possible so that he obtains a top up loan. He is however unsure of which option (repayment) to choose.

Additionally, Kaluma recently mobilised five other farmers (3 men and 2 women) to form an investment club. The bank guided that they would need a committee of 3 comprising at least 1 woman. If the possible number of ways of choosing the committee exceeds 5, Kaluma will volunteer to be the chairman, so that only 2 other members are chosen.

Task

Use mathematical evidence to help Mr. Kaluma make informed decisions.

Item 4

A community learning centre in a growing town plans to improve the sports culture in the community by constructing a sports centre where both indoor and outdoor games can be explored.

To raise funds for the project, the centre management plans to organize a sports fundraising dinner at a cost of 5,500,000. Community members and well-wishers are expected to pay a participation fee to attend the event. The total funds expected to be collected from the event, depend on the number of participants and are modeled by:

$F(x) = 200x^2 + 12{,}000x + 50{,}000$

Where;

- x is the number of participants attending the dinner.
- $F(x)$ is the total amount of money collected in (Uganda shillings).

The management has agreed that the fundraising dinner will only be considered successful, if the money collected is at least equal to the cost of organizing the event.

In addition, the management has launched a sports centre infrastructure fund to support the construction of a swimming pool.

The management contributes money to the fund every 18 months, in the 1st month, they contribute Ugx 1,200,000 and the monthly contributions increasing by Ugx 50,000 each subsequent month.

After 18 months, the accumulated money is to be invested in a bank that offers 10% compound interest per annum.

The management estimates that Ugx 40,000,000 will be required to complete the project. They have agreed that construction will only begin when the investment fund reaches or exceeds this amount.

Task

Use mathematical justifications to advise the management.

PART II

Respond to **one** *item from this part.*

Item 5

A certain company that packages different types of drinks introduced a new product in a community. The company management wanted to know whether people liked the product. They carried out a survey by asking 30 random community members to rate the product on a scale of 1-5. The information was summarised in the table 1 below;

table 1

Scale	1	2	3	4	5
No. of people	2	8	13	6	2

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If the variance is greater than 1, the sale of the new product would be halted.
The management further wants to compare the sales at the depot over the past 9 months shown in the table 2 below, on a quarterly basis.

table 2

Month	1	2	3	4	5	6	7	8	9
Sales(In millions)	70	82	40	70	62	90	45	76	65

If the estimated sales in the 10th term are greater than 90 million shillings, the management will consider increasing the wages of their workers. Additionally the depot is officially supported to be opened four days a week. However, some residents have complained that its only opened twice a week. A quick survey reveal the following information in table 3 below.

table 3

x	1	2	3	4
$P(X = x)$	0.15	0.35	0.41	0.08

Where;

- x represents the number of times the depot is open.
- $P(X = x)$ represents the respective probabilities. If the expected number of times the depot is open is less than 3, the depot supervisor would be changed.

Some residents also complained about a particular soft drink that would make them drowsy. The probability that the drink could make someone drowsy was 0.25. A sample of 20 residents was investigated. If the probability that at least 8 of them get drowsy is greater than 0.5, the drink would be removed from the shelves.

Task
Basing on numerical evidence, help the company management to make appropriate decisions.

Item 6
The National volley ball championships are scheduled for this term. According to the organisers the weight of the learners is normally distributed with an average of 50 kg and a variation of 1. The sports department in your school has 50 learners to choose from randomly, and would like to take at most 12 students of only learners whose weight exceed 49 kg, otherwise they will not participate. They need your help on the feedback to give the organisers.
Additionally a group of 10 learners would like to move with the team as cheer leaders. The sports department has requested for the details of their ages and mass, which are shown in table 4 below;

table 4

Age (yrs)	12	13	11	12	14	13	14	13	12	15
Weight (kg)	40	35	37	43	45	40	36	42	38	47

The department will consider taking the group, if the relationship between their age and weight is very strong.

In the past, the learners had complained about the poor feeding by organizers during the National games, rating them as follows on a scale of 1-5.
5 rated the food 1, 13 rated it 2, 14 rated it 3, 8 rated it 4 and 20 rated it 5.
The department would consider special meals for the learners during the championship if the variability exceeds 1.2.

Task
Using numerical evidence, help the sports department to make informed decisions.

END

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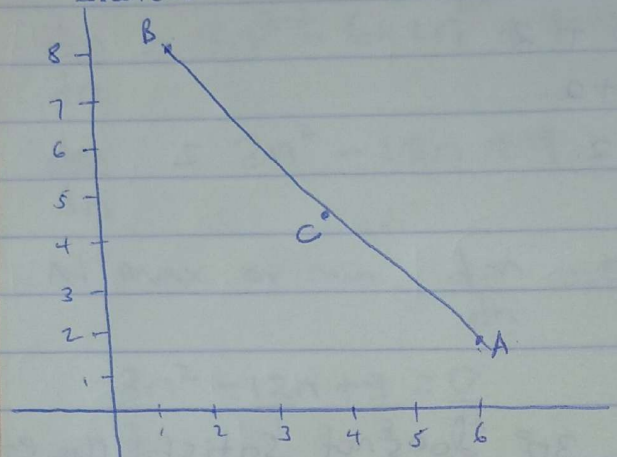
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WAKISHA SUBMATH 2026 PROPOSED GUIDE.

ITEM ONE

$OA = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$ $OB = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$ $OC = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$

Sketch.



If points lie on the same line, they are parallel to each other

$AB = OB - OA$

$= \begin{pmatrix} 2 \\ 8 \end{pmatrix} - \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$

$AC = OC - OA$

$\begin{pmatrix} 4 \\ 5 \end{pmatrix} - \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$

Ratio. $\frac{AB}{AC} = \frac{x_1}{x_2} = \frac{-4}{-2} = 2.$

$\frac{AB}{AC} = \frac{y_1}{y_2} = \frac{6}{3} = 2.$

$\frac{AB}{AC} = 2.$

$AB = 2AC$

Since the ratio is the same, then they are parallel to each other ie AB is a scalar multiple of AC.

He should not relocate pole C because it is in line with A and B.

(b) $3\cos 2\theta - \sin\theta + 2 = 0$

For Angle of Incline $\theta = 30°$

LHS

$3\cos(2\times30) - \sin 30° + 2$

$3\cos 60 - \sin 30 + 2$

$3\times0.5 - 0.5 + 2$

$1.5 - 0.5 + 2$

$1 + 2$

$= 3$

$3 \neq 0$

The angle of Incline 30° does not satisfy the equ, he should find another alternative.

$3\cos 2\theta - \sin\theta + 2 = 0$

$\cos 2\theta = 1 - 2\sin^2\theta$

$3(1 - 2\sin^2\theta) - \sin\theta + 2 = 0$

$3 - 6\sin^2\theta - \sin\theta + 2 = 0$

$-6\sin^2\theta - \sin\theta + 5 = 0$

$6\sin^2\theta + \sin\theta - 5 = 0$

$6\sin^2\theta + 6\sin\theta - 5\sin\theta - 5 = 0$

$6\sin\theta(\sin\theta + 1) - 5(\sin\theta + 1) = 0$

$(6\sin\theta - 5)(\sin\theta + 1) = 0$

Factors of 30

$-5 \quad 6$

either

$6\sin\theta - 5 = 0$

$\sin\theta = 5/6$

$\theta = \sin^{-1}(5/6)$

$\theta = 56.44°$

or

$\sin\theta + 1 = 0$

$\sin\theta = -1$

$\theta = \sin^{-1}(-1)$

$\theta = -90°$

He should Incline the plane at an angle of 56.44° for Vehicle Inspection.

ITEM TWO

$M = n^3 - 6n^2 + 9n + 2$.

For maximum out put, $\frac{dM}{dn} = 0$

$\frac{dm}{dn} = 3n^{3-2} - 6 \times 2n^{2-1} + 9n^{1-1} + 0$

$\frac{dm}{dn} = 3n^2 - 12n + 9$.

At max or min, $\frac{dm}{dn} = 0$

$\frac{3n^2}{3} - \frac{12n}{3} + \frac{9}{3} = \frac{0}{3}$

$n^2 - 4n + 3 = 0$

$n^2 - n - 3n + 3 = 0$

$n(n-1) - 3(n-1) = 0$

$(n-3)(n-1) = 0$

either $n - 3 = 0$ or $n - 1 = 0$

$n = 3$ $n = 1$

Second derivative.

$\frac{d^2M}{dn^2} = 6n - 12$.

when $n = 3$.

$\frac{d^2m}{dn^2} = 6(3) - 12$

$= 6 > 0$.

$n = 3$ is a minimum number of days.

for $n = 1$

$\frac{d^2m}{dn^2} = 6(1) - 12$

$= -6 < 0$

$n = 1$ is a maximum number of days.

So when $n = 1$,

$M = n^3 - 6n^2 + 9n + 2$.

$M = (1)^3 - 6(1)^2 + 9(1) + 2$

$M = 6$ tonnes.

The school should consider selling part of the produce because 6 tonnes are greater than 5 tonnes.

$\frac{ds}{dt} \propto 2t$

$ds = k(2t)\,dt$

$s = 2kt$

Initially $t = 0$, $s = 1$ tonne.

$s = 2kt$

$1 = 2$

$\frac{ds}{dt} \propto 2t$

$ds = k(2t)\,dt$

$\int ds = \int 2kt\,dt$

$s = \frac{2kt^2}{2} + c$

Initially, $t = 0$, $s = 1$

$1 = k(0)^2 + c$

$c = 1$

$s = kt^2 + 1$

After $t = 3$, $s = 5$ tonnes.

$5 = k(3)^2 + 1$

$9k = 4$

$k = 4/9$

$$s = \frac{4}{9}t^2 + 1$$

when $t = 8$

$s = \frac{4}{9}(8)^2 + 1$

$s = 29.444$ tonnes.

The school should widen its customers base beyond its immediate community because projected sales of 29.4 tonnes by 8 months are far less than the 150 tonnes target.

ITEM THREE.

Option one is an Arithmetic Progression with first term a, Common difference, d.

$U_n = a+(n-1)d$.

$U_5 = 90,000$.

$90000 = a+(5-1)d$.

$a+4d = 90000$ ---①

$U_{12} = 160,000$.

$160000 = a+(12-1)d$

$a+11d = 160,000$ ---②

$$\begin{array}{r} a+11d = 160000 \\ -\ a+4d = 90000 \\ \hline 7d = 70000 \end{array}$$

$d = 10000$

$a+4d = 90000$

$a+4(10000) = 90000$

$a = 90000 - 40000$

$a = 50000$.

Sum of 3million, required is time n,

$S_n = \frac{n}{2}(2a+(n-1)d)$

$3000,000 = \frac{n}{2}(2\times50000 + (n-1)10000)$

$6000,000 = n(100000 + 10000n - 10000)$

$6000000 = n(10000n + 90000)$

$\frac{10000n^2}{10000} + \frac{90000n}{10000} - \frac{6000,000}{10000} = \frac{0}{10000}$.

$n^2 + 9n - 600 = 0$.

$n = \frac{-9 \pm \sqrt{9^2 - 4(1)(-600)}}{2\times1}$

$n = -29.404$ or $n = 20.404$

∴ time required is 21 months.

Option 2.

$7^x = 12$.

Introduce log both sides.

$\log 7^x = \log 12$.

$x = \frac{\log 12}{\log 7}$.

$x = 1.27698.9$

x = 2 months (approximately)

He should use option two. because it takes 2 months shorter compared to option one of 21 months.

3M	2W	Total
2	1	3
1	2	3.

$^3C_2 \times {}^2C_1 \quad + {}^3C_1 \times {}^2C_2.$

$6 + 3.$

$= 9$ ways

Kaluma should volunteer to be the chairman since 9 ways exceed 5.

ITEM FOUR

Cost = 5,500,000

$F(x) = 200x^2 + 12000x + 50000$

For the event to be successfull, Money collected must be greater than Cost Incured. ie

$F(x) \geq$ Cost

$200x^2 + 12000x + 50000 \geq 5,500,000$

$\frac{200x^2}{200} + \frac{12000x}{200} + \frac{-5450000}{200} \geq \frac{0}{200}$

$x^2 + 60x - 27250 \geq 0$

$x^2 + 60x - 27250 \geq 0$

For Value of x,

$x^2 + 60x - 27250 = 0$

$x = \frac{-60 \pm \sqrt{60^2 - 4(1)(27250)}}{2 \times 1}$

$x = -197.7796$ or $x = 137.7796$

$\therefore x = 138$ participants.

Funds raised by 138 participants

$F(138) = 200(138)^2 + 12000(138) + 50000$

$=$

First month, $a_1 = 1,200,000$ $\quad d = 50,000$

$S_{18} = \frac{18}{2}(1200,000 \times 2 + (18-1) \times 50,000)$

$S_{18} = 9(2400,000 + 850000)$

$= 9 \times 3250000$

Money after 18 months $= 29250000$

$P = 29250000$, $r = 10\%$

$A = P(1 + \%/100)^t$

$A = 29250000\,(1 + 19/100)^t \geq 40,000,000$

$A =$

$(1.1)^t \geq \frac{40,000,000}{29250000}$

$(1.1)^t \geq 1.36752$

Introduce log both sides:

$\log(1.1)^t \geq \log 1.36752$

$t \geq \frac{\log 1.36752}{\log 1.1}$

$t \geq 3.284$

$t = 4$ years

For $t = 3$yrs,

$29250000(1.1)^3 = 38931750$

for $t = 4$

$29250000(1.1)^4 = 42824925$

The construction will only begin after 4 years.
because thats when the amount $42824925 > 40M$

ITEM FIVE.

Scale x	No of People f	fx	fx^2
1	2	2	2
2	8	16	32
3	13	39	117
4	6	24	96
5	2	10	50
	$\Sigma f = 31$	$\Sigma fx = 91$	$\Sigma fx^2 = 297$

$$\text{Var}(x) = \frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2$$

$$\frac{297}{31} - \left(\frac{91}{31}\right)^2$$

$$= 0.9636$$

The sale of the new product will not be halted because Variance is $0.9636 < 1$

Sales	Moving total	
70		
82	262	65.5
40	254	63.5
70	262	65.5
62	267	66.75
90	273	68.25
45	276	69
76		
65		

On graph.

UGANDA NATIONAL EXAMINATIONS BOARD

(to be fastened together with other answers to paper)

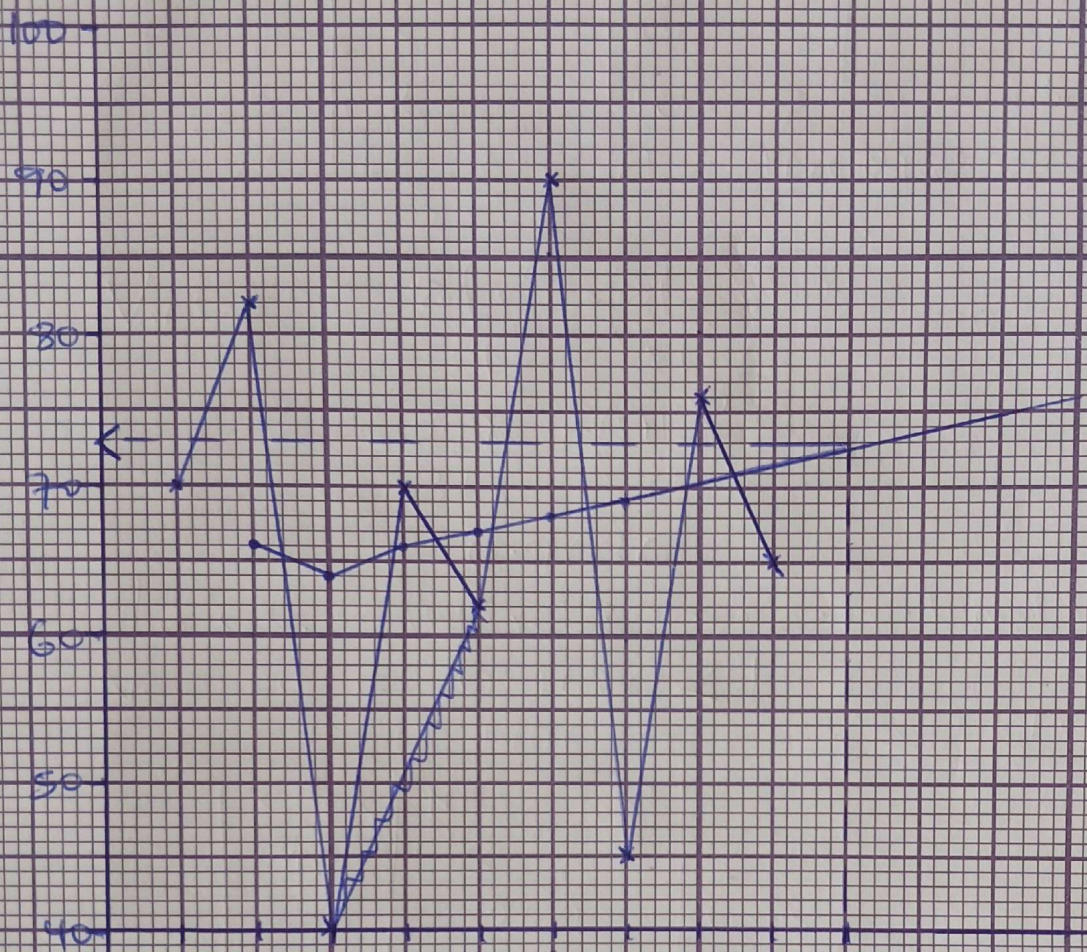
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Moving Average = 73

$$\frac{45+76+65+x}{4} = 73$$

$x = 292 - 186$

$x = 106$ millions

The management will consider increasing the wages of their workers because estimated sales were 106 millions greater than the 90millions

x	1	2	3	4
$P(x=x)$	0.15	0.35	0.41	0.08
$xP(x=x)$	0.15	0.7	1.23	0.32

Expectation, $E(x) = \sum_{All\,x} xP(x=x)$

$0.15 + 0.7 + 1.23 + 0.23$

$E(x) = 2.4$

The depot supervisor will be changed because Expected number of times the depot is open is 2.4 which is less than 3.

Let x represent making someone drowsy.

P - success probability q - failure probability.

$P = 0.25$ $q = 1 - 0.25$

$n = 20$ $q = 0.75$

$P(x \geq 8) = P(x=8) + P(x=9) + P(x=10) + P(x=11) + P(x=12) + P(x=13) + P(x=14) + P(x=15) + P(x=16) + P(x=17) + P(x=18) + P(x=19) + P(x=20)$

Using $P(x=x) = {}^{n}C_{r}\, p^{r} q^{n-r}$

$= 0.06089 + 0.02706 + 0.0099 + 0.00301 + 0.00075 + 0.00015 + 0.00003 + 0.00000 + \ldots\ldots$

$= 0.10179$ $= 0.1018$

OR

$P(x \geq 8) = 1 - P(x \leq 7)$

$= 1 - (0.00317 + 0.02114 + 0.06694 + 0.13389 + 0.18968 + 0.20233 + 0.10861 + 0.1124)$

$= 1 - 0.89818$

$= 0.1018$

The drink will not be removed from the shelves because the probability of atleast 8 people are drowsy is $0.1018 < 0.5$.

ITEM SIX

Normal distribution

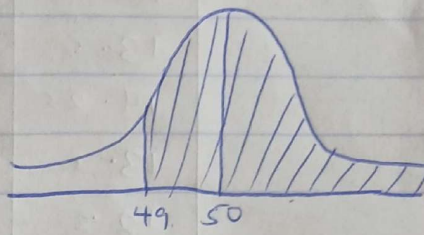
let x be the weight of learners.

$x \sim N(\mu\ \delta^2)$

$\mu = 50$ kg Var(x) $\delta^2 = 1$ $\delta = 1$

$$Z = \frac{x - \mu}{\delta}$$

~~$P(x \leq 12)$~~ $P(x > 49)$.



$$P\left(Z > \frac{49 - 50}{1}\right)$$

$P(Z > -1)$

$P(0 < Z < 1)$

$0.5 + 0.3413$

$= 0.8413$.

Using Spearman's Rank Correlation,

Age (yrs)	Weight	R_x	R_y	$d = R_x - R_y$	d^2
12 8	40	8	5.5	2.5	6.25
13.5	35	5	10	-5	25
11 10	37	10	9	2	4
12 8	43	8	3	5	25
14 2.5	45	2.5	2	0.5	0.25
13.5	40	5	5.5	-0.5	0.25
14 2.5	36	2.5	9	-6.5	42.25
13 5	42	5	4	1	1
12 8	38	8	7	-1	1
15 1	47	1	1	0	0
				Σd^2	105

$$\rho_c = 1 - \frac{6\Sigma d^2}{n(n^2-1)}$$

$$= 1 - \frac{6 \times 105}{10(10^2-1)}$$

$$= 1 - \frac{630}{990}$$

$$= 0.3636$$

The relationship btn their age and weight is very weak, not very strong, therefore the department will not consider taking them.

1	2	3	4	5
5	13	14	8	20

f	x	fx	fx^2
5	1	5	5
13	2	26	52
14	3	42	126
8	4	32	128
20	5	100	500
$\Sigma f = 60$		$\Sigma fx = 205$	$\Sigma fx = 811$

$$\text{Variance} = \frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2$$

$$= \left(\frac{811}{15}\right) - \left(\frac{205}{15}\right)^2$$

$$= 27.0222$$

$$\text{Variance} = \left(\frac{811}{60}\right) - \left(\frac{205}{60}\right)^2$$

$$= 1.8431$$

The department should consider special meals for learners during championship since 1.84 is greater than 1.2.